

Bicycle



Figure: Rendered Picture of the CAD Model of a bicycle.

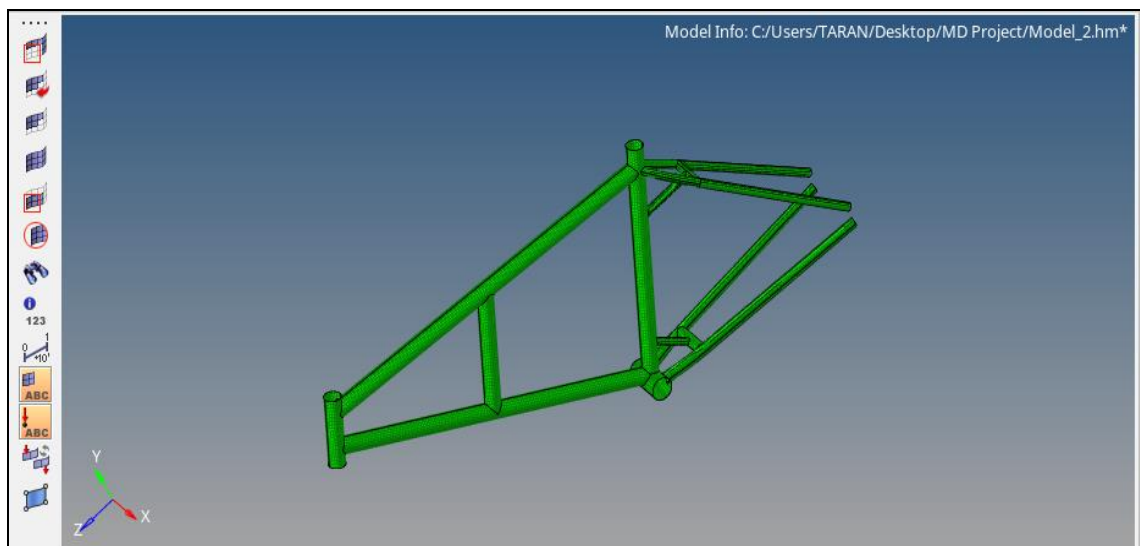
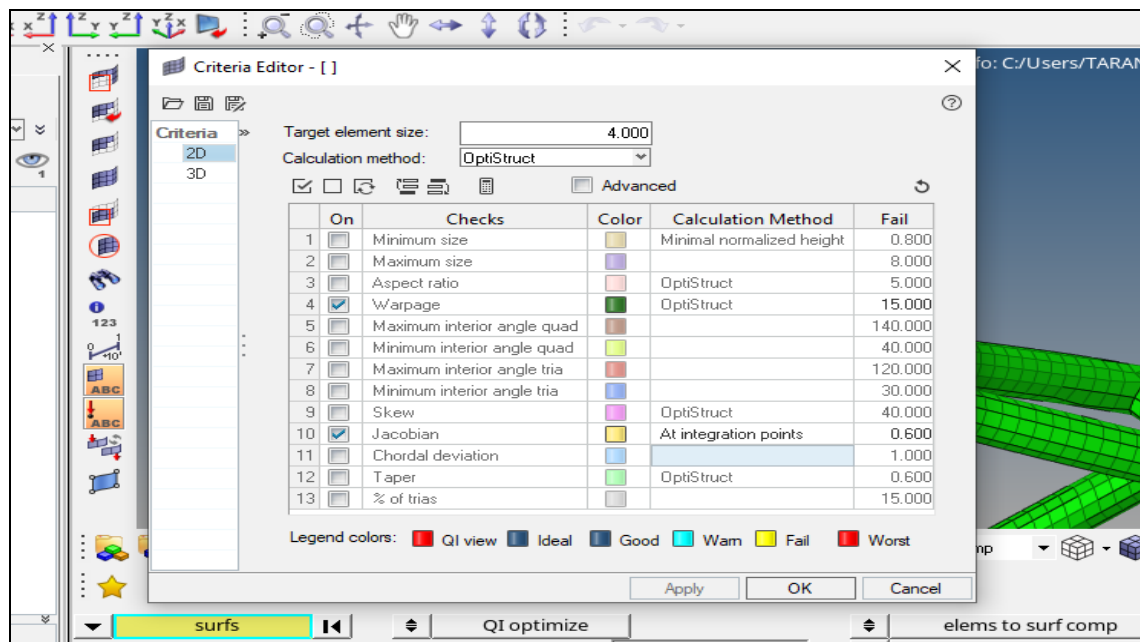
Introduction:

- This report consists of the Design and analysis of a bicycle.
- The design created was to be validated experimentally. Due to time and resource constraints, physical testing was not a feasible option and thus the use of computer simulation was found to be a more viable solution.
- Finite Element Analysis or **FEA** was used for validation of the component, it being a widely used tool for industrial applications. This method divides a complex structure into a finite number of elements.

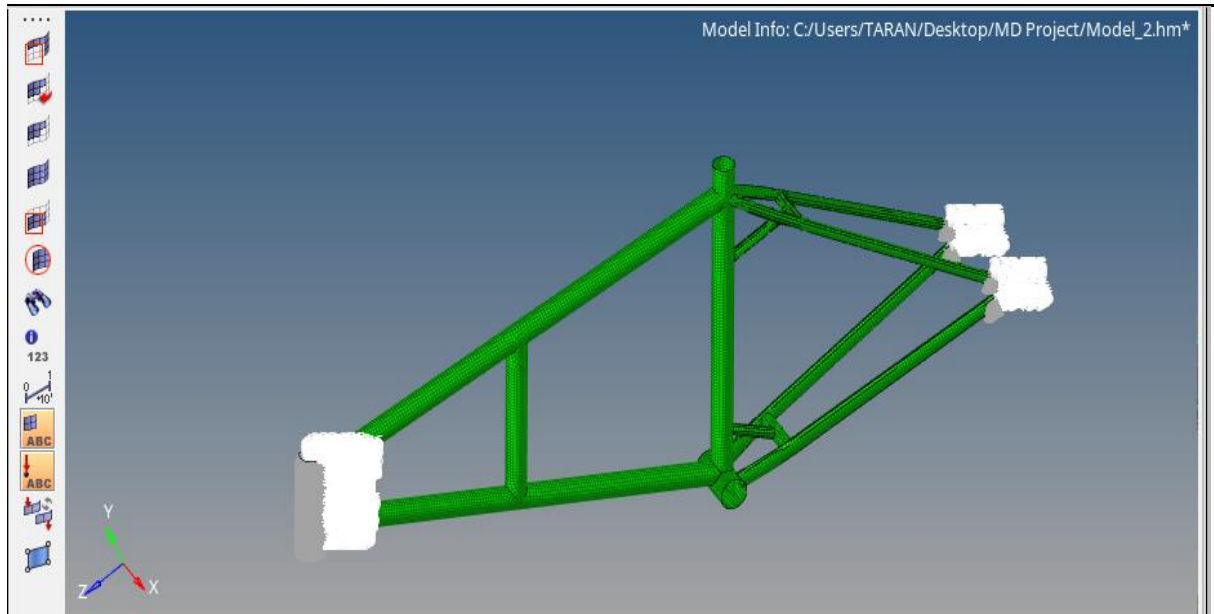
- The elements have corner points called nodes, which represent their degrees of freedom. The force applied on one node is used to calculate the displacement of the node using basic principles of strength of materials and the solution is interpolated across the entire domain.
- Thus, an approximate solution is obtained through this method. The software used was HyperMesh and the solver used was OptiStruct, due to its ability to effectively solve implicit analysis problems.
- Implicit or static analysis of a structure is to ensure the component is safe against loads in the elastic region (where Hooke's Law is obeyed in the stress strain curve of a material). This provides the foundation for further complex explicit or dynamic analysis.

Analysis of Bicycle Chassis:

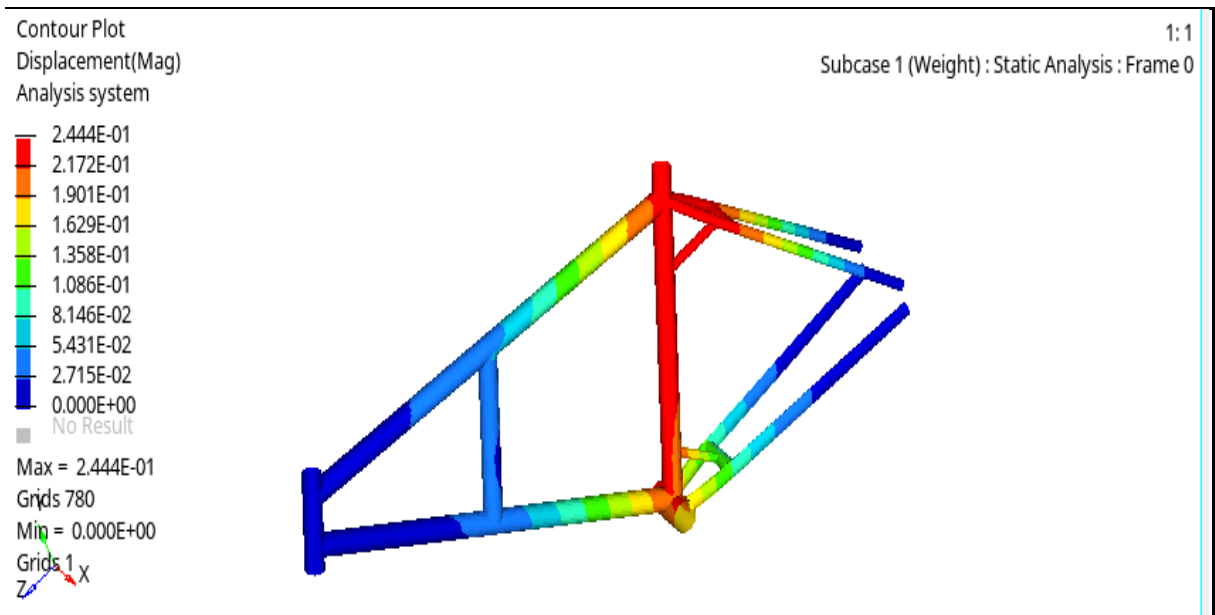
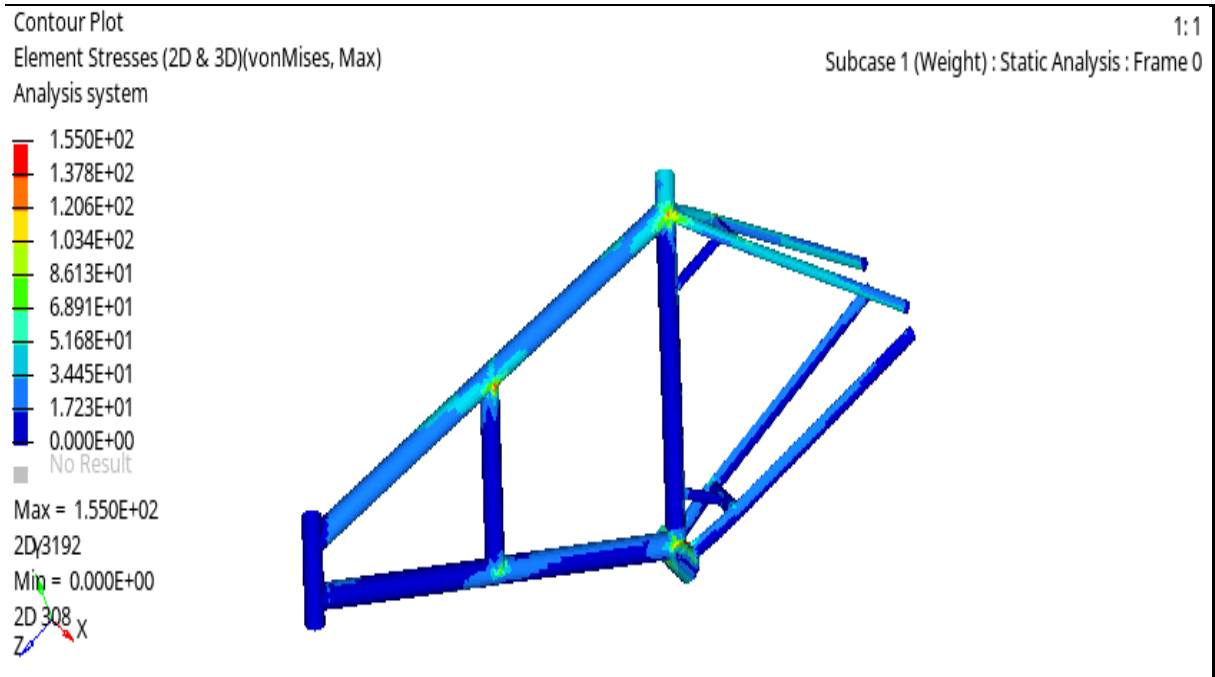
1. The geometry was imported in IGES format (standardization of CAD model for further processing). The geometry cleanup was performed and mid-surface was created for the geometry for uniform thickness of tubes. The tubes' thickness was chosen to be 1.6 mm. The material chosen was AISI 4130 due to its higher strength to weight ratio and higher bending strength and stiffness. The tubes were assigned the respective properties.
2. Analysis of a complex component through FEA requires discretization of the component into smaller elements known as **meshing**, because an object has infinite degrees of freedom and meshing divides it into a finite number of nodes, for simplicity of calculations. The mesh element size was chosen to be 4 mm and the Jacobian was set to 0.6 and Warpage was set to 15 degrees. The mesh was optimized further, after error elimination and the structure was ready for analysis.



3. Constraints were added to the end points, representing areas stationary with respect to the structure, or in general, the **hard points** of the structure were kept stationary.



4. The force was applied at the seat mounting position. Load applied was 500 kg, which was equal to 5000 N of force acting downwards, distributed equally among selected nodes.
5. The load step was created for the constraints and the load and the analysis was carried out.
6. The maximum stress induced in the cycle was 155 MPa and the maximum deformation was 0.2444 mm. Thus, the FOS (Factor of Safety) was calculated to be 5.09. Thus, the structure was safe.



Selection of the Tyres:

- Considering ease of manufacture and cost-effectiveness while still prioritizing durability and performance for daily use, a solid choice would

be tires made from a high-quality butyl rubber compound with a Kevlar puncture-resistant belt.

- Butyl rubber is a common material used in bicycle tires due to its excellent air retention properties and resistance to punctures. It's also relatively easy to manufacture, which helps keep costs down.
- The addition of a Kevlar belt provides an extra layer of protection against punctures without significantly increasing production complexity or cost.
- Kevlar is lightweight, durable, and offers excellent puncture resistance, making it ideal for daily commuting where avoiding flats is essential.
- By combining these materials, you get a tire that strikes a good balance between durability, puncture resistance, ease of manufacture, and cost-effectiveness, making it well-suited for daily use